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A steady-state wavefield polarity-aware U-Net for ultrasonic guided waves depth-resolved imaging

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ABSTRACT

The laser scanning full wavefield capture (LSFWC) technique enables rapid acquisition of two-dimensional maps of local velocity variations of ultrasonic guided waves (UGWs), thereby facilitating quantitative spatial visualization of damage in thin-walled structures. However, this technique faces a trade-off among spatial resolution, the accuracy of damage estimation, and detection efficiency: the spatial sampling rate influences the limit of velocity estimation, whereas increasing the sampling rate renders the full wavefield data acquisition extremely time-consuming. To tackle these difficulties, we propose a U-Net based imaging approach to reconstruct high-quality depth-resolved images from steady-state wavefield polarity (SSWP) of spatial sub-Nyquist sampled UGWs. The developed SSWP-aware U-Net can automatically segment the wavefield, extract local phase information, and perform encoding and decoding to establish approximate nonlinear mapping between wavefield and spatial distribution of UGWs velocity variations. The selection of SSWP as input features greatly accelerates the acquisition of the training dataset, and further enhances the robustness and generalization ability of U-Net trained on simulated dataset. We experimentally compared the proposed imaging method to wavenumber estimation methods. The results demonstrated that the proposed SSWP-aware U-Net is significantly superior to traditional methods in terms of image reconstruction quality and defect depth resolution especially with local spatial sub-Nyquist sampling.

1. Introduction

With the rapid advancement of modern industrial society, the demand for the reliability and integrity assessment of materials, components, and structures has been continuously increasing, thereby imposing higher requirements on nondestructive testing and evaluation (NDT&E) as well as structural health monitoring (SHM) technologies [1–3]. As a specialized modality of ultrasonic testing (UT), ultrasonic guided waves (UGWs) inspection not only retains the advantages of strong penetration, high sensitivity to volumetric defects, low cost, and safety for human operators, but also enables long-range propagation and the inspection of entire cross-section of waveguide structures through a single excitation, thereby demonstrating significant potential for engineering applications and considerable research value [4–6]. Leveraging features such as amplitude variation [7], phase deviation [8], time of flight (ToF) [9], reflection and transmission ratios [10], energy attenuation [11], and mode conversion [12] of damage-scattered waves, UGWs locating/imaging techniques (including phased array [13], sparse array [14], and tomography [15]) have been developed. Nevertheless, accurately characterizing the severity of damage remains a significant challenge.

In contrast to contact-based UGWs transduction methods, laser scanning full wavefield capture (LSFWC) technique offers flexible and high-precision measurement of the full surface wavefield, enabling detailed observation of wavefield anomalies arising from damage-wave interactions [16–18]. This capability makes it an indispensable tool for in-depth investigations into the coupling mechanisms between structural damage and UGWs propagation. By exploiting the dispersive characteristics of UGWs, where the wavenumber of specific modes varies with the frequency-thickness product [19], depth-resolved detection of defects can be achieved based on laser-scanned full wavefield measurements. A critical step toward this goal is the extraction of

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the two-dimensional wavenumber field from the measured full wavefield. Among the available approaches, Instantaneous Wavenumber Estimation (IWE) and Local Wavenumber Estimation (LWE) are currently the most widely adopted techniques for characterizing the spatial distribution of wavenumber associated with UGWs propagation [20].

The IWE algorithm estimates the local wavenumber by calculating the two-dimensional spatial phase gradient of the full wavefield [21–23]. Although computationally simple and efficient, making it suitable for real-time signal processing, it is sensitive to phase distortions caused by wave interference, noise, or defect scattering. These distortions can compromise defect detection reliability and hinder the precise reconstruction of the wavenumber field. Consequently, the IWE method places greater demands on the Signal-to-Noise Ratio (SNR) and anti-interference of the laser vibrometer instrument as well as the stability of the measurement environment. To address these limitations, recent studies have proposed various strategies to improve the performance of the IWE algorithm. These approaches, such as multi-shot [23] and multi-frequency [24] fusion method, have achieved notable improvements in noise robustness and defect detection accuracy, demonstrating the potential for enhancing the reliability and applicability of IWE-based methods in practical structural health monitoring scenarios. While the LWE method extracts the local wavenumber by applying a three-dimensional Short-Space Fourier Transform (SSFT) to the full wavefield data [25–27]. The introduction of spatial window significantly enhances the robustness of the wavenumber spectrum against noise and mode interference. However, it also introduces a fundamental trade-off between spatial resolution and wavenumber resolution, making it difficult to achieve both high spatial and high wavenumber resolution simultaneously. In addition, the LWE method requires the spatial window to be moved point-by-point across the laser scanning area to compute the local wavenumber spectrum, leading to substantial computational workload and extended data processing times.

However, the laser scanning interval inherently determines the wavenumber range of all spatial wavenumber estimation algorithms. When the local UGWs velocity decreases to a critical threshold (the spatial sampling interval becomes larger than twice the wavelength), the local wavenumber extracted from the full wavefield data becomes distorted. While reducing the scanning interval can yield a higher-quality spatial wavenumber spectrum, it undoubtedly decreases the efficiency of the LSFWC process. An alternative approach is to employ compressed sensing (CS) or deep learning super-resolution (DLSR) methods for the high-resolution reconstruction of full wavefield data with spatial sub-Nyquist sampling.

CS represents a signal or image in a sparse form using an appropriate basis, such as Fourier transform, cosine transform, or wavelet transform [28,29]. By exploiting sparsity and optimization methods for solving underdetermined equations, LSFWC data can be reconstructed at spatial sub-Nyquist sampling rates. Mesnil and Ruzzene [30] utilized a set of sparse measurement signals to identify the locations of excitation sources and scattering sources. Di Ianni et al. [31] investigated various basis functions in compressed sensing to reduce the time of LSFWC process. Wang et al. [32] designed a comprehensive dictionary incorporating a variety of waveforms generated under different conditions, enabling the decomposition of received signals in the spatial domain for damage localization, and employed a norm-based optimization algorithm to obtain a sparse solution corresponding to the physical location of the damage. However, the accuracy of the reconstruction is significantly influenced by the elements of the dictionary matrix, which is defined by the propagation equations of UGWs. Therefore, the acquisition of precise dispersion curves of UGWs in the structure is required a priori [33]. Additionally, factors such as the environmental temperature, and the potential interaction mechanisms between damage and UGWs can introduce uncertainty and distortion into the propagation equations [34]. Consequently, the dependence on the dictionary matrix and its sensitivity to uncontrollable factors may pose challenges to achieving robust and accurate wavefield reconstruction using CS algorithms.

Alternatively, deep learning offers an effective means to directly model the complex relationship between full guided wavefield and structural states by leveraging its automatic feature extraction and powerful nonlinear representation capabilities [35,36]. Ijjeh et al. [33] demonstrated DLSR method to reconstruct the high-resolution full wavefield from LSFWC data with spatial sub-Nyquist sampling. Their results indicated that DLSR outperforms CS in wavefield reconstruction accuracy, particularly in scenarios involving wave propagation through layered or heterogeneous regions. Ji et al. [17,34] successively proposed two deep learning architectures designed to accurately reconstruct the full wavefield and its spatial gradients from highly sparse measurements. These studies showcased the capability of deep learning to effectively process sub-Nyquist sampled wavefield data and achieve high-resolution damage imaging. However, DLSR approaches also exhibit notable limitations. The performance of deep learning models is highly dependent on the quality and diversity of the training dataset. Acquiring a large number of experimental samples is often time-consuming and cost-prohibitive. Although simulation-based data generation offers an alternative, the transient time-domain computations required to obtain full wavefield data are computationally intensive [37]. Furthermore, models trained solely on simulated dataset may suffer from limited robustness and generalizability when applied to LSFWC data, due to discrepancies between simulated and experimental conditions.

In this article, we propose a steady-state wavefield polarity (SSWP) -aware U-Net to reconstruct high-quality depth-resolved images from LSFWC data with spatial sub-Nyquist sampling. The SSWP-aware U-Net architecture consists of a symmetric encoder-decoder framework with skip connections and takes the SSWP of full guided wavefield as input, which enables to automatically model nonlinear interactions between the inspected medium and propagating UGWs, thus, establishing the mapping function linking the LSFWC data to the spatial distribution of the thickness within the inspected object. The SSWP of training dataset is directly generated by finite element analysis (FEA) in the frequency domain and the SSWP of LSFWC data is extracted from the direct wavefield, which is obtained through reference wavefield-based filtering. The performance of SSWP-aware U-Net is validated using LSFWC data from several aluminum alloy plate specimens with artificial thinning defects of various shapes, stepped and linear depth. The image quality of SSWP-aware U-Net is compared against the conventional wavenumber estimation methods to demonstrate the superior performance of proposed SSWP-aware U-Net on the image reconstruction quality and defect depth resolution especially with local spatial sub-Nyquist sampling.

2. Methodology

2.1. SSWP extraction

In FEA, the steady-state solution of the full guided wavefield can be directly obtained through frequency-domain computation. However, to prevent the superposition of reflected and incident waves in the complex amplitude spectrum, perfectly matched layer (PML) must be introduced to absorb outward-propagating waves, thereby eliminating boundary reflections and minimizing their interference within the computational domain. In practical measurements, boundary reflections of closed structures are inherent due to the closed nature of the structure. To extract the steady-state wavefield from LSFWC data, it is also necessary to eliminate the boundary reflection waves. One approach involves the use of damping materials to create absorbing boundaries; however, this method is relatively complex to implement and requires careful selection of materials and boundary configurations tailored to different modes and frequencies of UGWs. A more flexible alternative is to

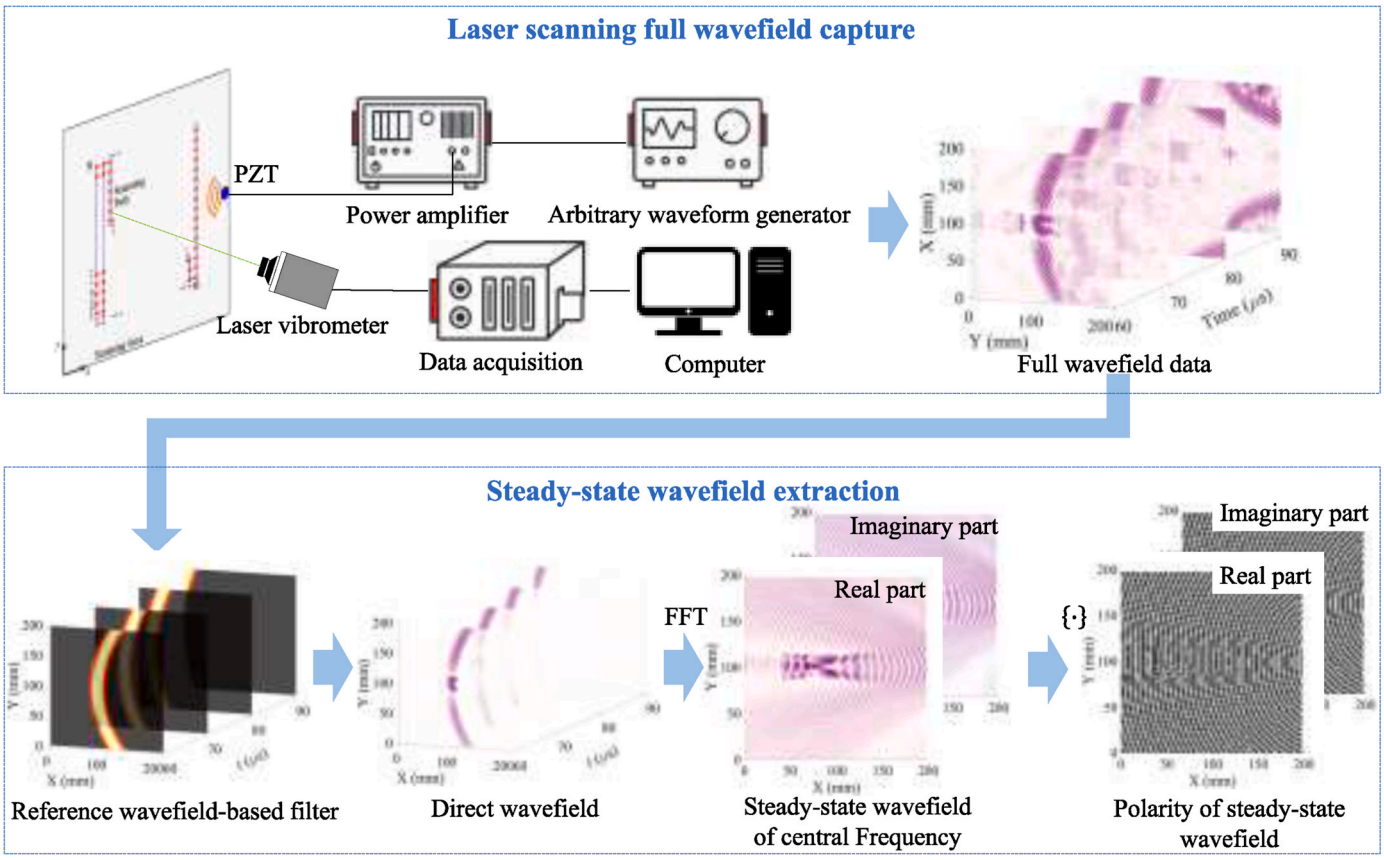


Fig. 1. The procedures of laser scanning full wavefield capture and steady-state wavefield extraction.

develop signal post-processing techniques based on filtering algorithms.

For a given structure, the theoretical guided wavefield can be computed using various analytical or numerical methods. This theoretical wavefield serves as the basis for constructing a reference wavefield-based filter. In the case of a typical isotropic plate-like structure, the propagation of UGWs can be described by the Navier equation, which governs elastic wave motion in solid media

$$(\lambda+\mu)\nabla(\nabla\cdot\boldsymbol{u})+\mu\nabla^2\boldsymbol{u}=\rho\frac{\partial^2\boldsymbol{u}}{\partial t^2} \tag{1}$$

where the boundary conditions of UGWs in plate-like structure are $\sigma_{zz}=\sigma_{xz}=\sigma_{yz}\equiv 0|_{z=\pm h}$, and

$$\boldsymbol{u}=\begin{cases} U_x(z)e^{[i(k_xx+k_yy+k_zz-\omega t)]} \\ U_y(z)e^{[i(k_xx+k_yy+k_zz-\omega t)]} \\ U_z(z)e^{[i(k_xx+k_yy+k_zz-\omega t)]} \end{cases} \tag{2}$$

where $\{U_x,U_y,U_z\}$ are the amplitude distribution of the particle vibration in directions of x,y,z. ∇ is the Laplacian operator and λ and μ is the Lamé constants.

For full guided wavefield imaging, the direct wavefield inherently contains features indicative of structural damage. Therefore, reference wavefield filtering requires only the extraction of the direct wavefield component. To compute the reference wavefield $v_{ref}(x,y,t)$ while satisfying the requirement of retaining only the direct wavefield component, the ray approximation method provides an efficient solution framework for the wave equation. This approach relies on two fundamental assumptions: (1) guided waves propagate without boundary reflections, and (2) their local wavefronts exhibit plane-wave characteristics with energy confinement along predefined ray paths. Under these prescribed conditions, the reference wavefield can be formulated as:

$$v_{ref}(x,y,t)=\mathscr{F}^{-1}\left\{\mathscr{F}\{s(t)\}e^{-ik_c(f)\sqrt{(x-x_0)^2+(y-y_0)^2}}\right\} \tag{3}$$

where $\mathscr{F}\{\cdot\}$ and $\mathscr{F}^{-1}\{\cdot\}$ are the Fourier transform and the inverse Fourier transform, (x_0,y_0) are the excitation position of UGWs, $s(t)$ is the excitation signal, $k_c(f)$ is the dispersion relation of detecting UGWs. The filter for isolating direct wavefield can be constructed by envelope of the reference wavefield

$$W_R(x,y,t)=\left|v_{ref}(x,y,t)+i\mathscr{H}\{v_{ref}(x,y,t)\}\right| \tag{4}$$

where the $\mathscr{H}\{\cdot\}$ represents the Hilbert transform. By applying reference wavefield-based filter $W_R(x,y,t)$ on the raw LSFWC data $v(x,y,t)$, the direct wavefield without boundary reflections can be obtained

$$\tilde{v}(x,y,t)=v(x,y,t)W_R(x,y,t) \tag{5}$$

Then the direct wavefield in the spatio-temporal domain is transformed into the frequency-wavenumber domain by Fourier transform and the steady-state wavefield of central frequency f_c can be extracted

$$\tilde{V}(x,y,f_c)=\mathscr{F}\{\tilde{v}(x,y,t)\}|_{f_c} \tag{6}$$

Finally, the SSWP of UGWs is obtained by applying the sign function $Sign\{\cdot\}$ to the steady-state wavefield

$$\mathscr{P}(x,y,f_c)=Sign\{\tilde{V}(x,y,f_c)\} \tag{7}$$

where the notation $Sign\{\cdot\}$ denotes the sign function, which returns the sign of its argument (i.e., $+1$ for positive values, -1 for negative values, and 0 if the value is zero).

By eliminating amplitude-related components from the original wavefield, the extraction of the polarization feature reduces susceptibility to amplitude errors caused by environmental variations, surface

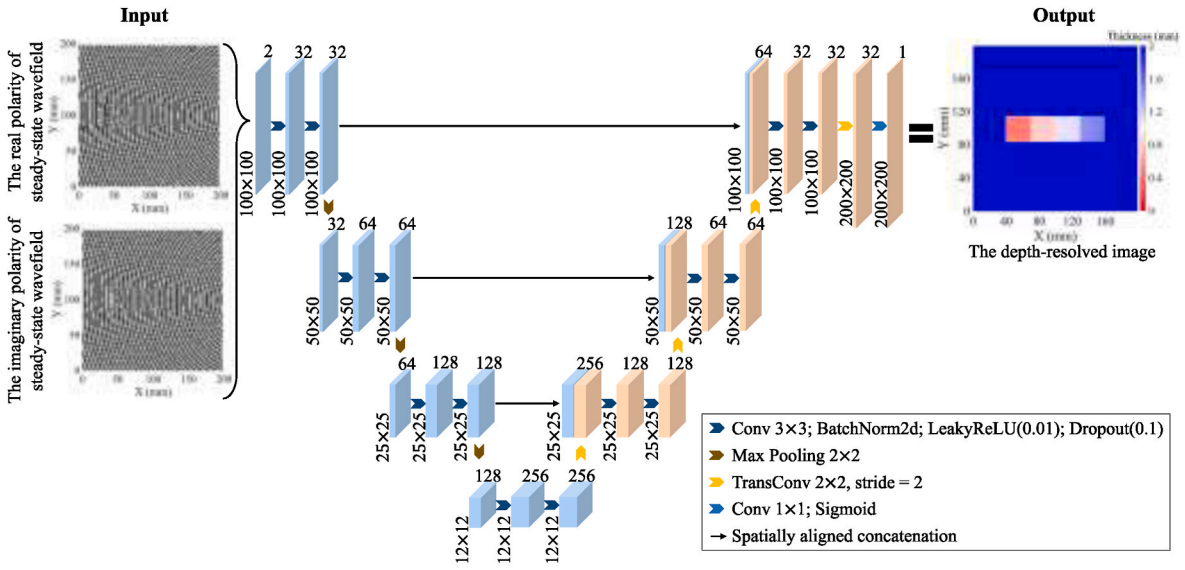


Fig. 2. The proposed SSWP-aware U-Net architecture.

irregularities, and measurement uncertainties, thus enhancing the robustness of the network input. Fig. 1 shows the procedures of LSFWC and steady-state wavefield extraction. It is noted that the SSWP contains both real and imaginary components, which are utilized as two separate input channels for the deep learning model.

2.2. Proposed SSWP-aware U-Net

The SSWP can be regarded as a typical two-dimensional image. U-Net, a well-established deep learning architecture for image processing, is particularly suitable for SSWP based imaging task due to its symmetric encoder-decoder structure and concatenation-based skip connections. This design allows for the preservation of rich spatial details, enabling automatic segmentation of the SSWP, extraction of local phase information, and the execution of encoding and decoding operations. Consequently, it facilitates the approximation of the nonlinear mapping between the guided wavefield and the spatial distribution of UGWs velocity variations. The architecture of proposed SSWP-aware U-Net is illustrated in Fig. 2.

The proposed SSWP-aware U-Net architecture is a three-level encoder-decoder structure with skip connections whose input is the SSWP of UGWs and output is the generated depth-resolved image. The contracting path at the left side of the network structure can be regarded as an encoder to perform feature extraction to capture the spatial phase

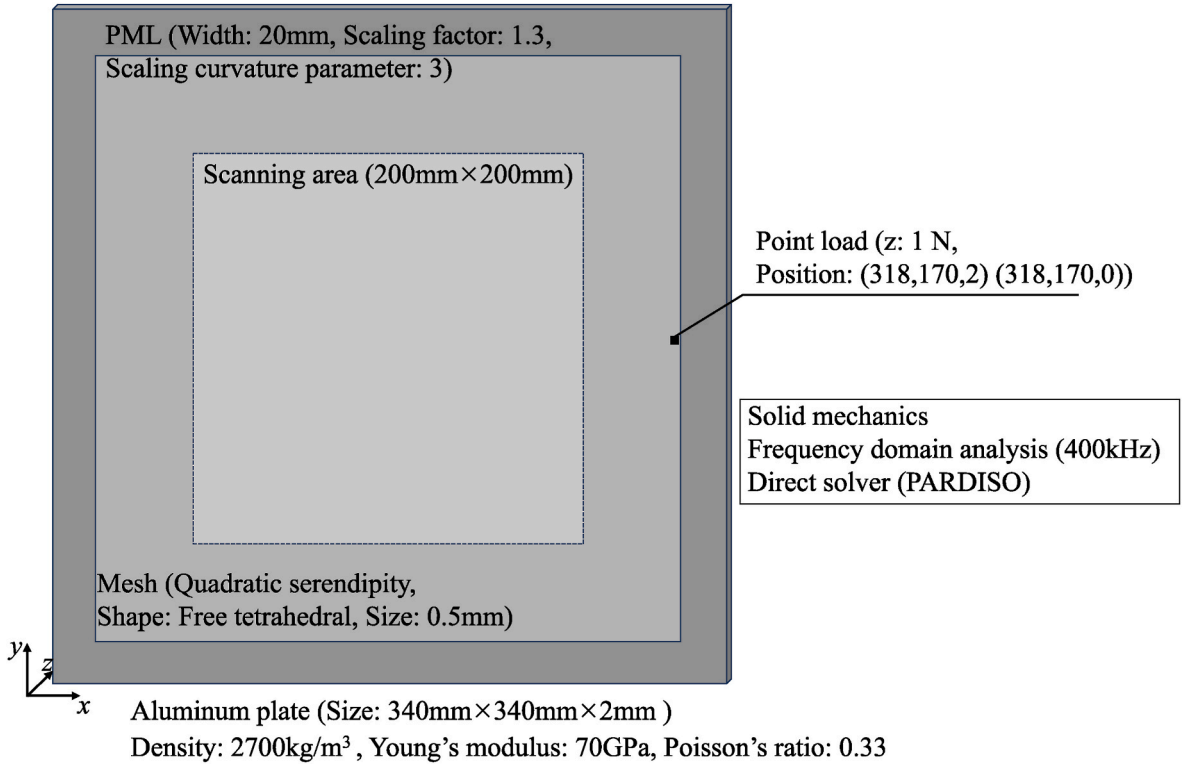


Fig. 3. Schematic diagrams of finite element simulation model.

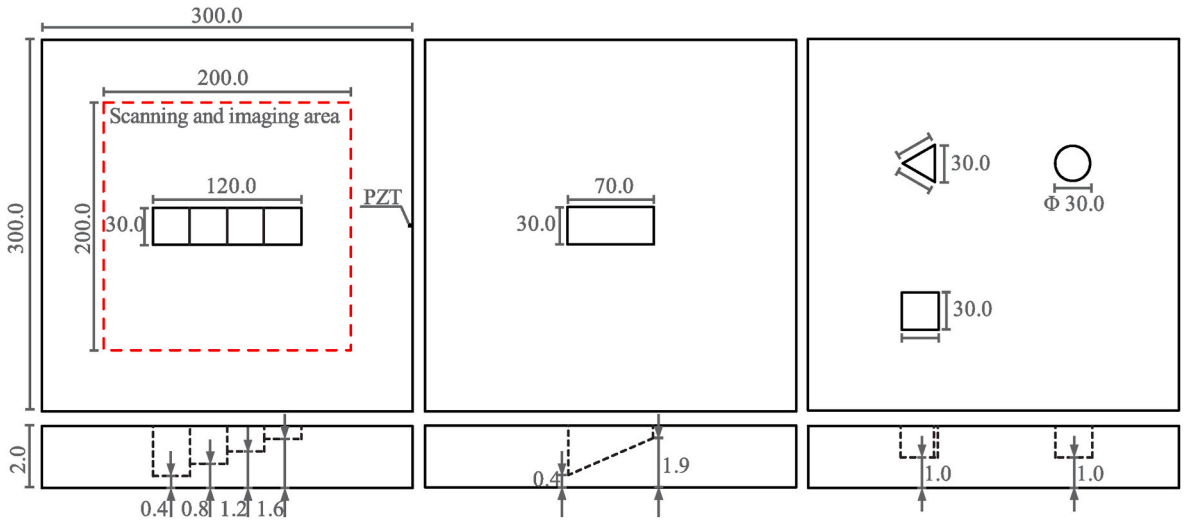


Fig. 4. Illustration of the bottom and side surfaces of the specimens (unit: mm).

distribution information in the SSWP data. The symmetric expanding path at the right side can be regarded as a decoder to generate the corresponding depth-resolved image. The size of input SSWP data is $100 \times 100 \times 2$, where 100 is the number of spatial sampling points of LSFWC in the directions of x and y, and 2 denotes the two input channels corresponding to the real and imaginary parts of the SSWP, respectively. Each encoding layer contains two 3×3 convolution kernels, each of which goes through a two-dimensional batch normalization layer and a leaky rectified linear unit (LeakyReLU), enabling the network to capture nonlinear features effectively and mitigate the risk of neuron inactivation in the negative input domain. Additionally, a dropout rate of 0.1 is applied after the activation function to prevent overfitting. Each encoding layer is then followed by a 2×2 max pooling layer for downsampling. The input SSWP data go through three downsampling layers, so the spatial dimension of the data tensor is contracted from 100×100 to 12×12. The decoding layer is similar to the encoding layer, but the max pooling is replaced by a 2×2 transposed convolution with a stride of 2, which gradually restores the resolution of the feature maps through the upsampling operation to reconstruct the depth-resolved image. In the final output layer, an additional upsampling operation is introduced using a 2×2 transposed convolution with a stride of 2, which increases the spatial resolution of the output to 200×200, thereby enhancing the imaging resolution. It is worth noting that, since the input to the SSWP-aware U-Net does not conform to standard image dimensions, dimensional mismatches may arise during the downsampling and upsampling operations, affecting spatially aligned concatenation. To address this, a tensor padding operation was applied after the first upsampling stage, expanding the feature map from 24×24 to 25×25. This ensures consistent spatial dimensions between corresponding layers in the encoder and decoder, preserving the structural symmetry of the network.

3. Implementation procedures

3.1. Dataset preparation

This section aims to demonstrate the training of a deep learning model capable of transforming LSFWC data into depth-resolved images. A supervised learning approach is employed to train the proposed SSWP-aware U-Net architecture, establishing a mapping between the SSWP of LSFWC data and the two-dimensional spatial distribution of the thickness of the inspected structure. However, obtaining large-scale experimental datasets that simultaneously contain LSFWC data and ground-truth thickness maps is both impractical and cost-prohibitive. Additionally, generating such datasets via time-domain FEA demands substantial computational resources and time. Therefore, the steady-state wavefield derived from the LSFWC data is adopted as the input feature for the network, as it can be efficiently obtained through frequency-domain FEA. To enhance the robustness and generalization capability of the U-Net trained on simulated dataset, the steady-state wavefield is further processed into the SSWP, which serves as the input feature to the network. The polarization operation removes amplitude-related information from the original wavefield, thereby mitigating errors in amplitude that may arise from environmental factors, surface roughness, or other measurement uncertainties in practical scenarios.

The steady-state wavefields of UGWs used for training data are generated using the solid mechanics module and frequency domain analysis in COMSOL Multiphysics 6.2. Fig. 3 shows the schematic diagrams of finite element simulation model. The simulation model consists of a $340mm \times 340mm \times 2mm$ aluminum plate with a density of 2700 kg/m^3, Young's modulus of 70 GPa, and a Poisson's ratio of 0.33, and no material damping. PMLs are implemented along all boundaries to absorb outgoing waves and eliminate reflections. Each PML has a width of 20 mm, with a scaling factor of 1.3 and a scaling curvature parameter of 3. The central $200mm \times 200mm$ region of the plate includes a series of randomly distributed defects with varying sizes and depths, which are generated using MATLAB. These defects are designed to simulate realistic thinning-type damage and serve as the target features within the imaging region. The A_0 mode of the UGWs was excited by applying point loads with identical orientation on both the upper and lower surfaces at (318, 170). For frequency-domain FEA, the guided waves can be generated using a single time-independent continuous force, with the effective excitation frequency determined by the frequency range computed by the solver. The simulation model is meshed using quadratic serendipity elements with shape of free tetrahedral and element size of $0.5mm$. The steady-state wavefield was computed using frequency-domain FEA with an implicit solver, with the PARDISO direct solver applied at 400 kHz to ensure numerical stability. The steady-state wavefields used for training are sampled at interval of 2 mm, yielding input data of size 100×100, whereas the ground-truth depth-resolved images are sampled at interval of 2 mm, resulting in label maps of size 200×200. The simulated dataset contains 6000 ground-truth depth-resolved images, along with their corresponding steady-state wavefield data. Each frequency-domain simulation requires approximately 160 seconds of computation on a platform configured with 288 GB RAM, Intel Xeon Silver 4210R CPU @ 2.40 GHz and NVIDIA RTX A6000 GPU. The total computational time for generating the entire dataset is

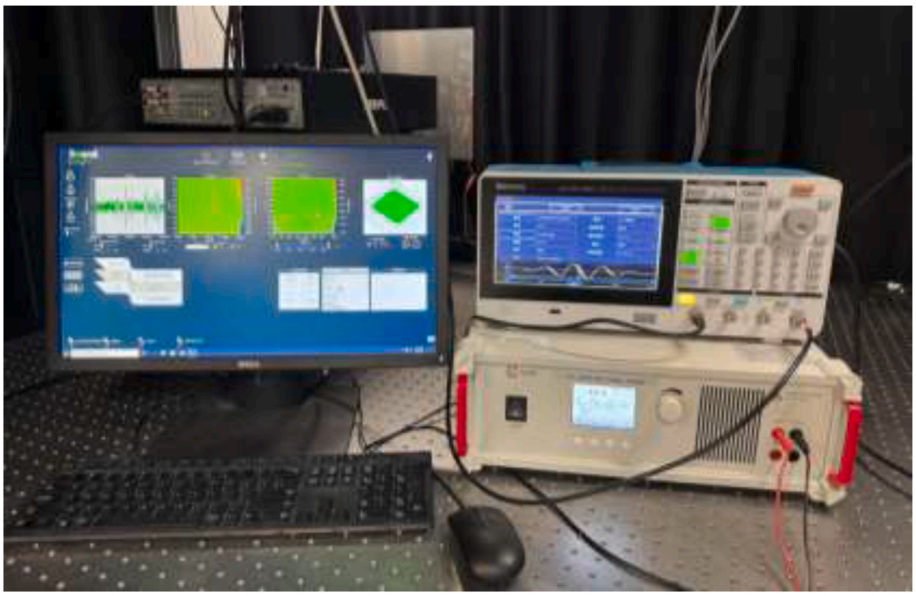


Fig. 5. Experiment configuration.

approximately 270 hours, which is substantially lower than that required for constructing time-domain simulated dataset of the same size [37].

3.2. Training strategy

The training of the proposed SSWP-aware U-Net is first conducted on a simulated dataset and subsequently validated on an experimental dataset. In the frequency-domain FEA, a total of 6000 SSWP samples are generated and used as inputs to the network, while the corresponding depth-resolved maps served as the target outputs. The entire set of simulated samples was randomly split into training and testing subsets with a ratio of 9:1, where the training set was used for model optimization and the testing set for performance evaluation. Final performance assessment of the trained network was carried out using the experimental LSFWC dataset. During training, the adaptive moment estimation with weight decay (AdamW) optimizer is employed with an initial learning rate of 0.001 and weight decay coefficient of 0.0001 to minimize the loss function. To prevent overfitting, a dropout strategy was adopted, and batch normalization was incorporated to mitigate the effects of parameter initialization and accelerate convergence. To compare the network performance, a U-Net model with the same architecture was also trained using the steady-state wavefield as input, under identical training strategy. All training procedures were performed on the NVIDIA RTX A6000 GPU.

3.3. Experimental setup

To validate the generalizability of proposed SSWP-aware U-Net on real UGWs data, experimental LSFWC data are acquired from aluminum plate specimens containing a series of artificially machined thinning defects with various shapes, stepped cross sections, and linear depth variations. The details about the bottom and side surfaces of the specimens were illustrated in Fig. 4. A schematic of the experimental setup is presented in Fig. 5. As shown in Fig. 5, UGWs are excited using a PZT-5H transducer with dimensions of $5mm \times 5mm \times 5mm$. The excitation signal consisted of a 5-cycle tone burst modulated by a Hanning window with a central frequency of 400 kHz. This signal was generated by Tektronix AFG31022 arbitrary waveform generator and subsequently amplified using Agitek ATA-2022B high-voltage amplifier. LSFWC data are acquired using Sound&Bright Tempo 2D two-wave mixing laser interferometric vibrometer. The specimen was mounted on an x-y scanning platform driven by two sets of stepper motors. The scanning platform was synchronously controlled via the trigger signal from the arbitrary waveform generator to enable precise spatial acquisition of the full wavefield. The temporal sampling rate is 25 MHz, and the spatial sampling interval is set to 2 mm.

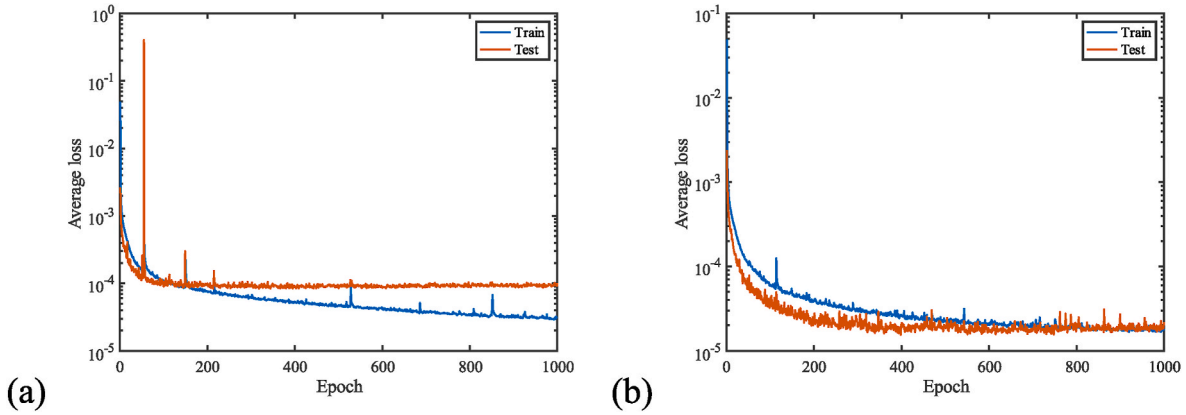


Fig. 6. Average loss of the network on the simulated dataset (a) SSWP-aware U-Net; (b) steady-state wavefield-aware U-Net.

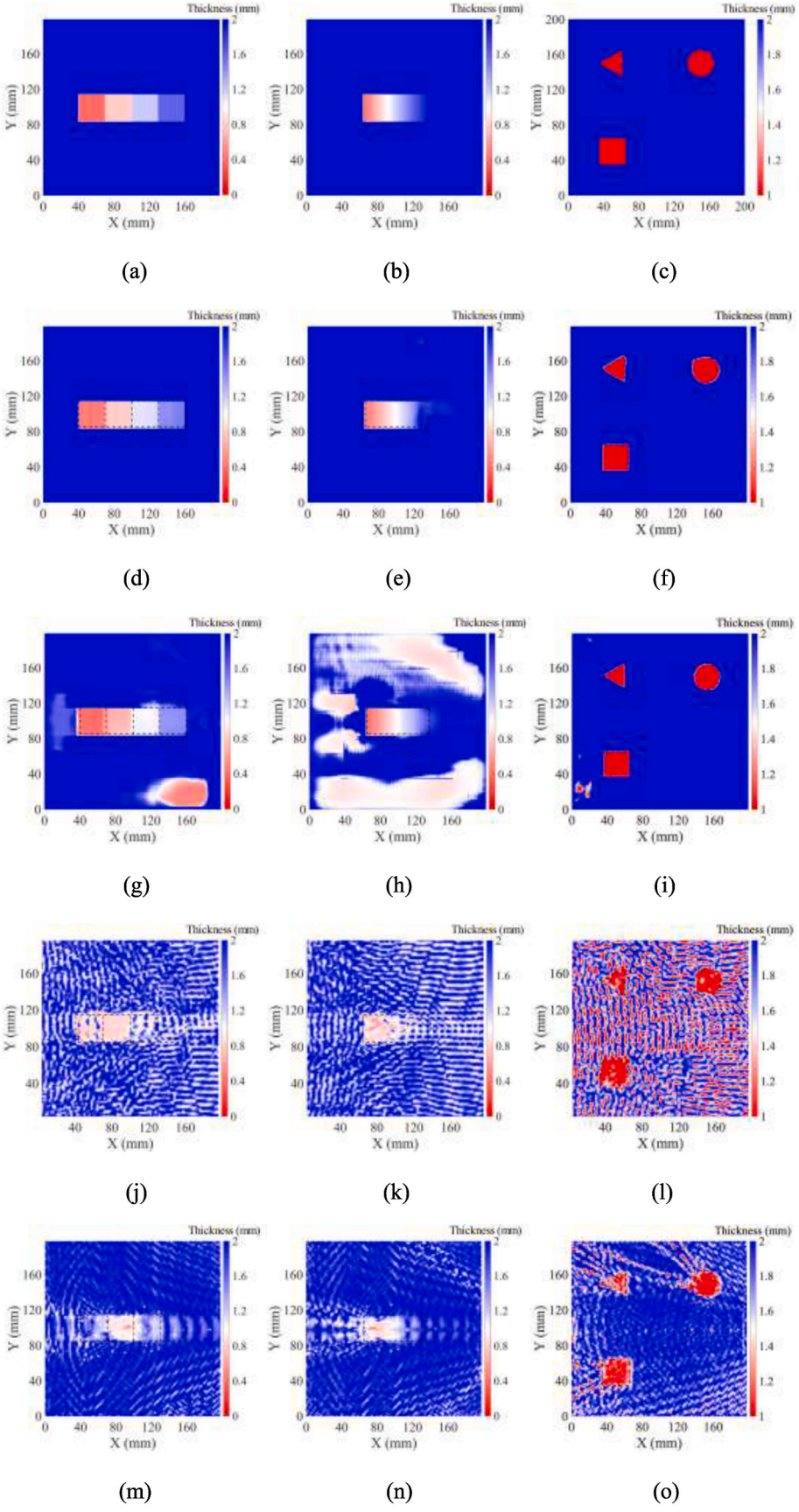


Fig. 7. (a–c) Real thickness distribution map of the specimens; Depth-resolved imaging results by (d–f) proposed SSWP-aware U-Net, (g–i) steady-state wavefield-aware U-Net, (j–l) LWE and (m–o) IWE. The dashed lines represent the boundaries of the defects.

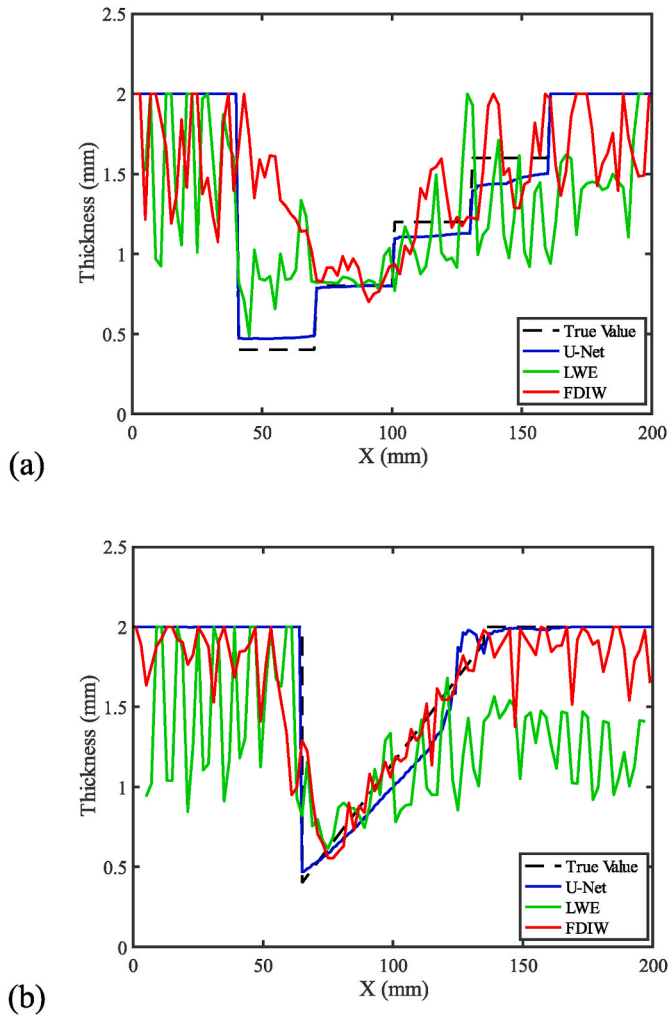


Fig. 8. Profiles from depth-resolved imaging results of specimens containing artificially machined thinning defects with (a) stepped cross sections and (b) linear depth variations, at the center of the vertical direction.

4. Results and discussion

As illustrated in Fig. 6, the average training and testing loss of SSWP-aware U-Net and steady-state wavefield-aware U-Net on the simulated dataset is monitored to evaluate model convergence and training stability. The loss curves indicate that both networks effectively learned the correlation between the wavefield and the spatial distribution of defect depth. Specifically, the SSWP-aware U-Net exhibits slight overfitting, as evidenced by a noticeable gap between the training and testing losses; however, both remained within the same order of magnitude. The steady-state wavefield-aware U-Net achieves marginally lower overall losses than the SSWP-aware U-Net, though the difference was not substantial, and the losses also remain within the same order of magnitude. The actual performance and generalization capability of the networks are further validated and compared against traditional wavenumber estimation-based imaging methods using experimental LSFWC data.

Fig. 7 shows the actual thickness distribution map of the specimens along with the imaging results obtained using the proposed SSWP-aware U-Net and other depth-resolved imaging methods, including steady-state wavefield-aware U-Net, LWE and IWE. It can be observed that the proposed SSWP-aware U-Net reconstructed images successfully capture the fine morphological features of the defects, with noise and artifacts effectively suppressed. This enables accurate assessment of both the shape and depth of each defect. These experimental results confirm the high-resolution depth-resolved imaging capability of the proposed deep learning-based approach. While the steady-state wavefield-aware U-Net with the same network architecture is also capable of accurately characterizing defects, its imaging results contain a considerable number of artifact regions. This can be attributed to the domain gap between the simulated and experimental data. In experimental conditions, factors such as environmental noise and variations in surface reflectivity of the specimen introduce additional disturbances to the amplitude distribution of the full wavefield. By leveraging only phase-related features through the SSWP input, the SSWP-aware U-Net is less sensitive to amplitude distortions caused by experimental uncertainties, thereby achieving more consistent and reliable imaging performance when transitioning from simulated to real LSFWC data. In traditional wavenumber estimation-based imaging methods, significant imaging noise is commonly observed. Moreover, according to the sampling theorem, the fixed spatial sampling interval of 2 *mm* in the experimental setup limits the maximum achievable depth resolution to approximately 1.2 *mm*. As a result, these methods tend to produce substantial depth estimation errors, particularly in regions containing deeper defects. To provide a clearer evaluation of the depth-resolved imaging performance, Fig. 8 presents depth profile cross-sections extracted along the vertical centerlines of two specimens with different thinning depth. The results demonstrate that the proposed SSWP-aware U-Net yields a significantly higher imaging signal-to-noise ratio compared to traditional methods and maintains reliable depth estimation performance even in areas with sub-Nyquist sampling, particularly in regions where the defect depth exceeds 1.2 *mm*. In summary, experimental results demonstrate that the proposed SSWP-aware U-Net exhibits strong cross-domain robustness and remarkably outperforms traditional wavenumber estimation methods in reconstructing high-quality depth-resolved images. In particular, it shows superior performance in artifact and noise suppression, as well as in overcoming the resolution limitations imposed by spatial sampling constraints.

5. Conclusion

In this work, we propose a deep learning-based approach aimed at directly reconstructing high-quality depth-resolved images from the SSWP of full guided wavefield data. Specifically, based on the two-dimensional UGWs propagation model, we design a reference wavefield-based filter that enables the extraction of boundary-reflection-free SSWP from LSFWC data. To replace conventional wavenumber estimation imaging algorithms, we develop an end-to-end SSWP-aware U-Net. Benefiting from the encoder-decoder architecture with skip connections, the proposed deep learning model is capable of implicitly capturing the complex nonlinear interactions between structural features and UGWs propagation, thereby learning an effective mapping between the spatial phase distribution of the steady-state wavefield and the spatial thickness distribution of the inspected structure. By using SSWP, which removes amplitude information and retains only phase-related features, as the input feature, the network achieves more consistent and robust imaging performance when transitioning from simulated data to real LSFWC data. Experimental results validate that the proposed SSWP-aware U-Net significantly outperforms traditional wavenumber estimation methods in reconstructing high-resolution depth-resolved images, particularly in terms of artifact and noise suppression and in overcoming resolution limits imposed by spatial sampling constraints. The proposed approach holds promise for a wide range of laser-scanned guided waves imaging applications.

CRediT authorship contribution statement

Xuan Li: Writing – original draft, Methodology, Conceptualization. **Lishuai Liu:** Writing – review & editing, Investigation, Funding acquisition. **Jiang Lu:** Visualization, Software, Methodology. **Yuncheng Zhang:** Supervision, Formal analysis, Data curation. **Dazhi Cong:**

Formal analysis, Data curation. **Yanxun Xiang:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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